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Challenges in the production of sustainable fuels from pyrolysis oil – Design of efficient catalysts for gasification of char

B. Matas Güell[a], G. van Rossum[b], W.P.M. van Swaaij[b], S.R.A. Kersten[b], L. Lefferts[a], K. Seshan[a,*]

[a] Catalytic Processes & Materials, Faculty of Science and Technology, University of Twente, P.O. Box 217, 7500 AE, Enschede, The Netherlands
[b] Thermo-Chemical Conversion of Biomass, Faculty of Science and Technology, University of Twente, P.O. Box 217, 7500 AE, Enschede, The Netherlands

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ABSTRACT

The catalytic performance of a ceria–zirconia based catalyst for (i) steam/CO_2 gasification and (ii) combustion of char produced during pyrolysis oil evaporation was investigated and compared with the non-catalytic scenario in the temperature range 600–800 °C. Kinetic studies revealed that the presence of the ceria–zirconia catalyst enhanced char gasification rates significantly, up to one order of magnitude at 800 °C, for both steam and CO_2 gasification. It is suggested that the red-ox properties of the ceria–zirconia catalyst play a significant role in char gasification, by allowing oxygen mobility and activation of H_2O and CO_2. Furthermore, XPS and SEM characterization studies showed that the catalyst also influences the nature of char formed during pyrolysis-oil evaporation to a great extent, by forming oxygenated species which are expected to be more reactive.

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1. Introduction

Biomass has recently received considerable attention as a potential sustainable feedstock for hydrogen production, as it would strongly contribute to the minimization of green house emissions. In the context of biomass, flash pyrolysis of lignocelluloses results in liquid pyrolysis oil. This makes it a convenient feedstock for storage, transport and processing as compared to the bulk biomass [1]. Catalytic steam reforming of pyrolysis oil is a promising option for hydrogen production. This reaction is carried out at high temperatures (>800 °C) [2]. Therefore, pyrolysis oil needs to be vaporized and heated to the desired reaction temperature. The high instability of pyrolysis oil at these high temperatures due to polymerization reactions results in carbonaceous solid residues, besides gases and vapors. These solid residues are referred to as homogeneous char. We have shown earlier and concluded [3] that formation of these carbonaceous deposits during pyrolysis oil evaporation cannot be avoided completely. These carbonaceous species can be detrimental to catalyst activity/stability and affect the efficiency of the overall reforming process [4].

Decoupling of (i) pyrolysis oil evaporation and (ii) reforming of the generated gases/vapors using a two-staged reactor concept (fluidized bed for vaporization + fixed bed for steam reforming) has been reported [5] to successfully improve steam reforming catalyst stability and improve the process. Contact between char and reforming catalyst can be avoided if char is retained and converted in the fluidized bed itself and a catalyst is essential for this [3]. Steam and CO_2 gasification are promising routes to enhance char removal during pyrolysis oil evaporation [3,6], producing at the same time a high value gas product (syngas). Pyrolysis oil is being evaporated at around 700 °C and at these low temperatures a catalyst is essential to achieve the required char gasification.

The envisioned down-stream processes determine the preference for steam or CO_2 as gasification agent. Steam gasification is used to produce a hydrogen-rich gas product. On the contrary, combined steam and CO_2 gasification is applied when the end application requires adjustment of the H_2/CO ratio to lower values (2–3), e.g., for methanol synthesis and Fischer–Tropsch process.

In the case that CO_2 and/or steam gasification of char are insufficient for the required levels of char removal, an external partial combustion step, e.g., *via* a circulating fluidized bed, is required. The heat released during this exothermic combustion can be integrated to (i) pyrolysis oil evaporation or (ii) the endothermic reforming step. The circulation of char (solid) at high pressures (10–30 bar) is the main challenge, as it results in difficult process operability.

There are parallels to be drawn between char and coal gasification, as the challenges are very similar. Exhaustive work on coal gasification [7,8] has shown that alkali metal compounds are effective catalysts for steam gasification of coal. Alkali metals are suggested to enhance gasification activity by assisting in the for-

* Corresponding author.
E-mail address: k.seshan@utwente.nl (K. Seshan).

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mation of OH groups [9,10]. In the well known Exxon Catalytic Coal Gasification (CCG) process, for instance, K_2CO_3 was used as catalyst. This allowed reaction temperatures as low as 650–750 °C. Similarly, enhanced gasification activity by alkali/alkaline earth based catalysts has been reported also for CO_2 gasification of coal [8,11,12]. In comparison to steam gasification, CO_2 gasification is carried out at higher temperatures even in the presence of a catalyst (750–900 °C).

The most widely reported coal gasification catalyst, K_2CO_3, however, presents quite a few problems. First of all, catalyst losses occur during gasification due to the formation of volatile KOH species. Secondly, interaction of catalyst with minerals present in coal leads to the formation of inactive and water insoluble compounds (e.g., potassium aluminum-silicates) from which recovery of the catalyst is difficult [11,12]. These catalyst losses make the gasification process uneconomic. Furthermore, it has been reported by Marshall and Smits [13] that potassium promotes the methanation reaction, which is favored at high pressure and low temperature (industrial desired conditions) and undesired in our case as part of the hydrogen would be consumed. Therefore, an alternative catalyst which has the ability to activate H_2O and CO_2 and avoiding the above mentioned problems is desired for minimizing the problems related to homogeneous char formation during pyrolysis oil gasification. Tomishige et al. [14] observed that CeO_2 supported catalysts, e.g., $Rh/CeO_2/SiO_2$, with the ability to activate both water and CO_2 [15,16], exhibited good activity in cellulose gasification. Ceria-based catalysts may thus be a good option for the steam and CO_2 gasification of char formed during pyrolysis oil evaporation.

The contact between catalyst and char is a crucial step in the gasification process, as it influences the catalytic activity to a large extent. In the case of coal gasification, a solution containing the catalyst precursor is sprayed on coal prior to gasification. In the case of pyrolysis oil, this approach cannot be applied since evaporation of pyrolysis oil is required previous to gasification. Instead, pyrolysis oil is sprayed on the catalyst particles in the gasification unit itself.

The aim of the present work is to develop an active and stable ceria catalyst for the (i) steam/CO_2 gasification and (ii) combustion of char formed during pyrolysis oil evaporation. Additionally, we aim at exploring and understanding whether the presence of catalyst, has any influence on the nature of char formed during the pyrolysis oil evaporation, which would in turn influence the reactivity/kinetics of gasification.

2. Experimental

2.1. Catalyst preparation

Ceria catalysts were prepared using the wet impregnation technique. Yttrium stabilized zirconia (TOSOH, TZ-8Y) was first calcined for 15 h at 850 °C (heating rate 5 °C min^{-1}) in flowing air (30 ml min^{-1}). The powder was then impregnated with an aqueous solution of $Ce(NO_3)_3{\cdot}6H_2O$ (Fuka) to yield a catalyst with 8.2% ceria loading. The $Ce(NO_3)_3{\cdot}6H_2O$ solution was mixed for 2 h and then dried at 50 °C in vacuum for 4 h in a rotary evaporator. Subsequently, the impregnated catalyst was first calcined at 700 °C (heating rate 5 °C min^{-1}) for 6 h and then crushed and sieved to give grains of 0.3–0.6 mm. This catalyst will be, hereafter, denoted as Ce–Zr–O.

2.2. Materials

The pyrolysis oil was produced in the Process Development Unit of VTT (Finland) from forest residue [17].

For non-catalytic experiments, finely powdered char, produced at 525 °C, was used. Char production is described in detail by van Rossum et al. in one of our earlier publications [3]. This char will be, hereafter, denoted as external char.

2.3. Contact between char and catalyst

The contact between char and catalyst will depend on the type of catalyst and the contacting efficiency in a fluidized bed. In order to verify the relevance of such contact, two extreme situations have been tested in this work: (i) absorption of the oil into the catalyst particles and subsequent heating to produce char *in situ* in the pores of the catalyst, resulting in optimal contact and (ii) physical mixing of char, produced externally, with the catalyst particles in a mortar leading to poor (interparticle) contact.

The preparation procedure for (i) was the following: approximately 40 mg of Ce–Zr–O or SiO_2 as an inert, reference material [18] (Aerosil 380, Degussa) was impregnated with pyrolysis oil (15–20 mg) in a small crucible. Subsequently, the impregnated catalyst was first heated up to 800 °C (10 °C min^{-1}) in flowing argon (60 ml min^{-1}) and then cooled down to room temperature under the same atmosphere. This char will be, hereafter, referred as internal char.

For (ii), finely powdered char (denoted as external char) was physically mixed manually in a mortar with known amounts of catalyst. This char will be, hereafter, denoted as physical mix char.

2.4. Catalyst and char characterizations

Catalyst composition was determined with X-ray fluorescence (XRF) spectroscopy (Phillips PW 1480 spectrometer). Char composition was determined with an elemental analyzer (Fisions Instruments 1108 EA CHN-S).

BET specific surface area and porosity were measured with nitrogen adsorption–desorption at 77 K in a Micromeritics TriStar instrument. The samples were first degassed at 300 °C in nitrogen.

X-ray photoelectron spectra (XPS) were recorded on a Physical Electronics Quantera XPS system using a monochromatic Al Kα (1486.6 eV) X-ray source operated at 24 W (beam diameter 100 μm) with charge neutralization. Survey scans were collected from −5 eV to 1345 eV with a pass energy equal to 224 eV and 0.8 eV step size. All scans were taken at a 45° tilt angle. The pressure inside the chamber remained $<2 \times 10^{-9}$ torr. Peak shape fitting was performed with Gaussian–Lorentzian sum function and Shirley background [19] subtraction with constraints applied to full width, half maximum (FWHM) values. All binding energies were referenced against a main C 1s peak at 285.0 eV.

Scanning electron microscope (SEM) images were obtained using a LEO 1550 instrument.

2.5. Reactivity measurements

CO_2 gasification experiments were performed in a Mettler Toledo thermo-gravimetric analyzer (TGA) equipped with good temperature control (±0.7 °C) and a balance (1 μg sensitivity). Experiments were performed isothermally in the temperature range 600–900 °C in a small crucible. Samples were heated rapidly (~150 °C min^{-1}) to the desired temperature in a mixture of carbon dioxide (20 ml min^{-1}) and argon (40 ml min^{-1}). All experiments were carried out until char conversion was completed or reached steady values. Additional to the TGA balance, the samples weight loss was quantified with a very accurate external balance (0.01 μg sensitivity).

Combustion experiments were performed in the same system. The samples were heated to 800 °C at 5 °C min^{-1} in a mixture of air (20 ml min^{-1}) and argon (40 ml min^{-1}). The initial and final weights of samples were also quantified with the same external balance. It is noteworthy to mention that isothermal experiments were not

done for combustion experiments as reaction would have occurred to a too large extent during heating up.

Steam gasification experiments were carried out in a fixed-bed reactor. A quartz tube (Ø 45 mm, length 400 mm) was used which was placed inside an electrically heated oven. A steam generator was used to create a constant steam flow (300 °C, 0.15–0.5 g/min) and preheated nitrogen (ca. 200 °C, 9 ml/min) was added as an internal standard. The amount of steam added compared to the char sample (ca. 4–10 mg) was high enough that zero conversion of steam could be assumed. Samples were placed at the end of the oven to ensure adequate preheating of the steam/nitrogen and isothermal gasification. The samples containing char only were pre-mixed with quartz (ca. 150 mg) to minimize the pressure drop which was created due to the fine structure of the char. However some pressure drop (0.2–0.5 bar) over the sample was observed. For the samples containing char and catalyst (9 mg of char + 55 mg Ce–Zr–O) no quartz was used. The maximum pressure drop observed in this case was 0.3 bar. The reactor outlet was cooled to ~15 °C and the steam was condensed out completely. A micro-GC (Varian CP-4900) was used to analyze the gas composition and the mass loss was calculated based on the determined gas composition.

2.6. Definitions and reaction equations

The degree of conversion, $X(t)$, is obtained in this study from the following expression,

$$X = \frac{m_0 - m_t}{m_0} \quad [-]$$

where m_t is the char mass at a certain time t and m_0 represents the char mass at the beginning of the gasification. The rate of gasification is here defined as the change of conversion over a time period, $r_{wt} = dX/dt$ [s^{-1}]. The rate of combustion is here defined as the change of conversion over a temperature range $r_{wT} = dX/dT$ [$°C^{-1}$].

The reaction equations involved in char gasification and combustion are listed in Eqs. (1)–(4)

$$C_xH_yO_{z(s)} + (x-z)H_2O \rightarrow xCO + \left(\frac{y}{2} + x - z\right)H_2$$

(Steam gasification) (1)

$$C_xH_yO_{z(s)} + (x-z)CO_2 \rightarrow (2x-z)CO + \left(\frac{y}{2}\right)H_2 \quad (CO_2 \text{ gasification}) \quad (2)$$

$$C_xH_yO_{z(s)} + \left(\frac{y}{4} + x - \frac{z}{2}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O \quad (\text{Combustion}) \quad (3)$$

$$CO + H_2O \leftrightarrow CO_2 + H_2 \quad (\text{Water–gas shift}) \quad (4)$$

3. Results

3.1. Catalyst characterization

The ceria content in the catalyst was 8.2 wt%, in the range intended for preparation. Specific surface areas for Ce–Zr–O and SiO_2 powders after calcination at 700 °C were 18 $m^2\,g^{-1}$ and 351 $m^2\,g^{-1}$, respectively. The nitrogen sorption isotherm and the pore volume distribution (inset) for the Ce–Zr–O catalyst are shown in Fig. 1. The isotherm, assigned to types II and IV [20], was typical for particles which either are nonporous or possess macropores and relatively low external surface areas. Fig. 1 (inset) also shows

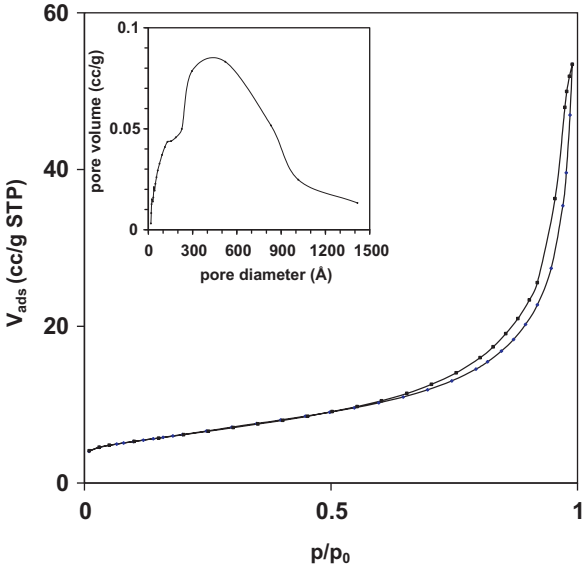


Fig. 1. Nitrogen adsorption isotherm and pore volume distribution (inset) of Ce–Zr–O after calcination at 700 °C.

that Ce–Zr–O presented a broad pore distribution, with a mean pore diameter ~50 nm.

3.2. Composition of pyrolysis oil and external char

The carbon and hydrogen content in the pyrolysis oil were 40.6 wt% and 7.6%, respectively. The rest (51.8 wt%) was mainly oxygen and small amounts of other components, e.g., sulfur, nitrogen and ash. Char produced externally consisted of 76.7% carbon, 3.4% hydrogen and 19.9% oxygen.

3.3. Characterization of internal char deposited on Ce–Zr–O

Fig. 2 illustrates char formed during pyrolysis oil evaporation in the presence of Ce–Zr–O. A typical SEM photograph of the cross-section of a catalyst particle with internal char is shown in Fig. 2a. A zoomed in image of the cross-section (Fig. 2b) reveals that char formed during pyrolysis oil evaporation was evenly distributed in the catalyst (dark spots are carbon), filling the catalyst pores. An energy dispersive X-ray spectroscopy (EDX) analysis of this section (Fig. 2c) confirms the presence of carbon and thus, char. Fig. 3a shows a typical tilted top-view SEM image of internal char deposited on a catalyst particle. EDX analysis (Fig. 3b) clearly indicates that these deposits were mainly composed of char. As clearly seen, char was not exclusively located in the catalyst pores. A small amount of char formed an external thin surface layer. As shown in Fig. 3c (cross-section), this external deposits layer was only few μm thick (up to 5 μm). Considering that catalyst particles size is in the range 300–600 μm, the volume of the external deposits layer is equivalent to a very small fraction of the volume of a catalyst particle (~2.5%).

Presence of internal char in Ce–Zr–O resulted in an increase in specific surface area (86 $m^2\,g^{-1}$) as compared to that determined in the absence of char (18 $m^2\,g^{-1}$). Nitrogen adsorption and pore volume distribution for internal char deposited on Ce–Zr–O are shown in Fig. 4. The physisorption isotherm, Type IV, is characteristic for materials possessing mesoporosity. The corresponding hysteresis loop was associated with narrow slit-like pores, Type H4. Decrease in macropore volume (compare insets in Figs. 1 and 4 (top)) when Ce–Zr–O was impregnated with internal char indicates that the char was located in the catalyst pores, influencing the textural properties

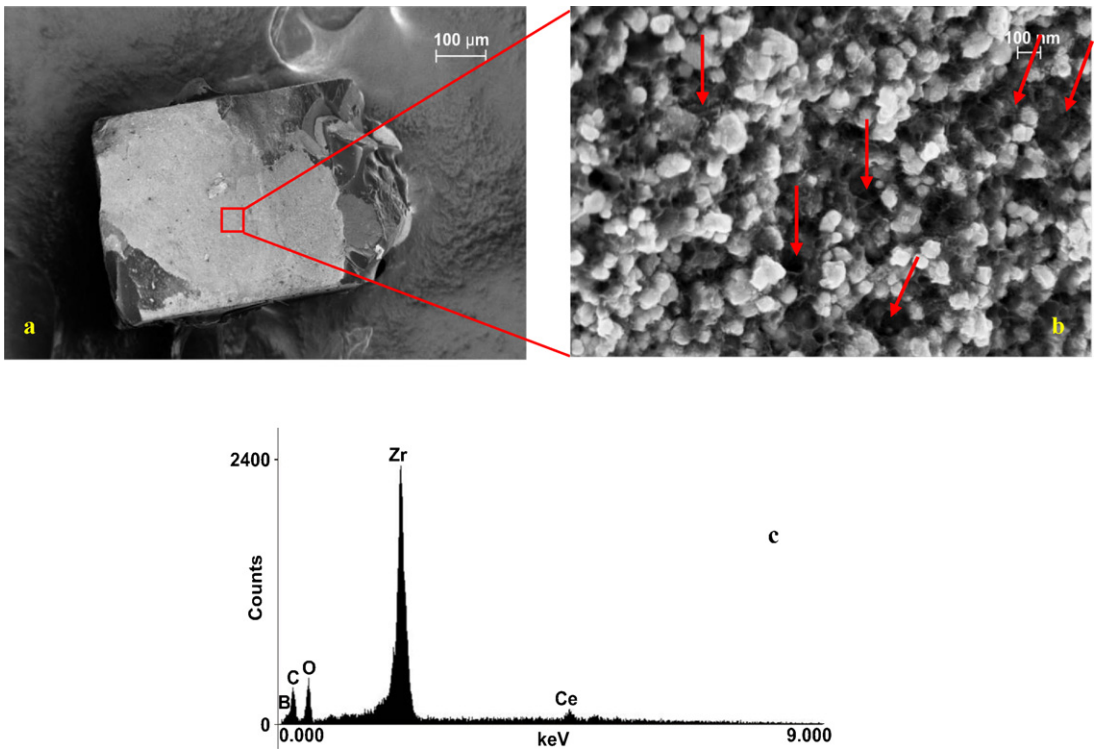


Fig. 2. Cross-sectional SEM images on a cutled catalyst particle of internal char deposited on Ce–Zr–O, (a) low magnification and (b) high magnification. EDX analysis of a selected area (c). The light area is the cross section.

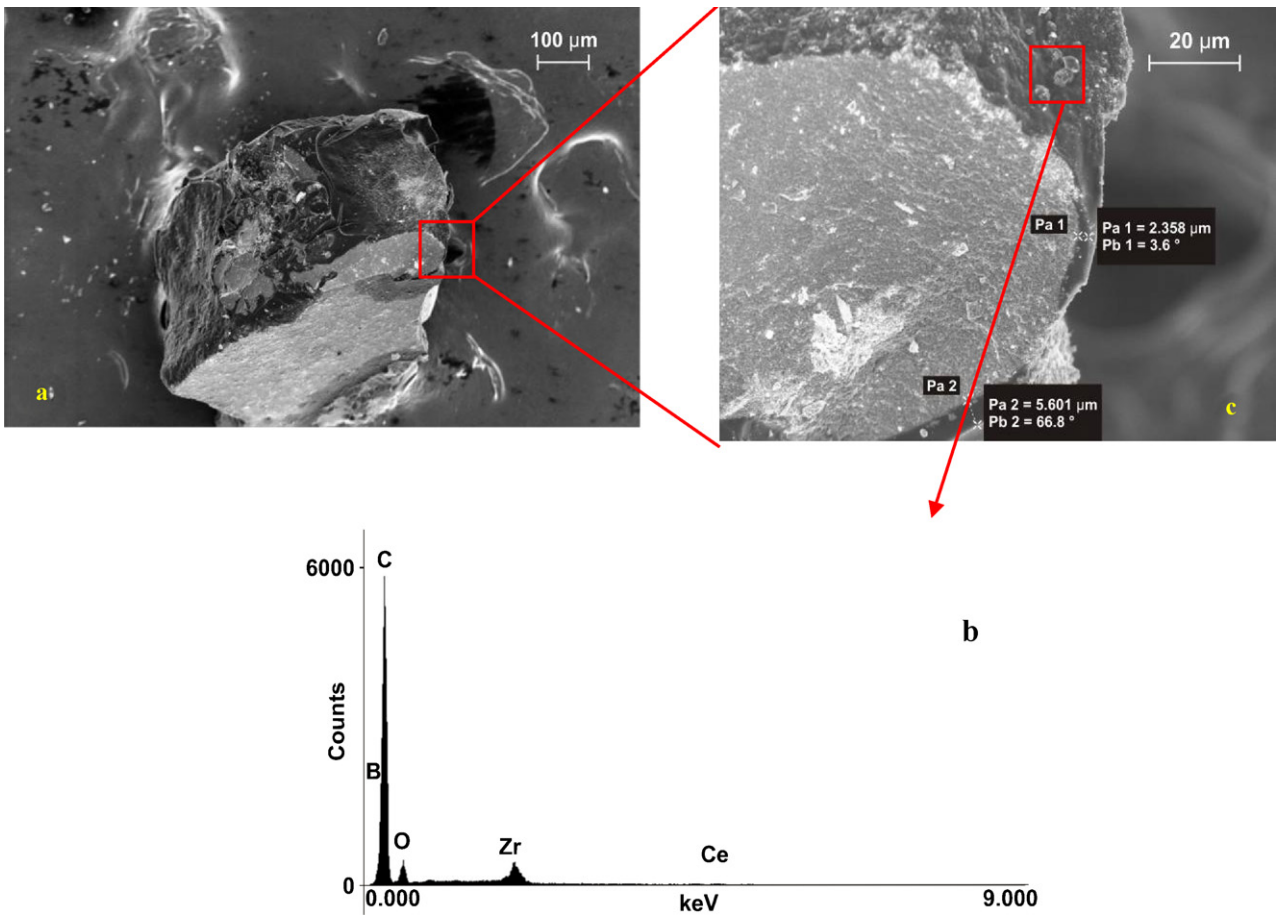


Fig. 3. Top-view SEM images of internal char deposited on Ce–Zr–O, (a) low magnification and (b) high magnification. EDX analysis of a selected area (c).

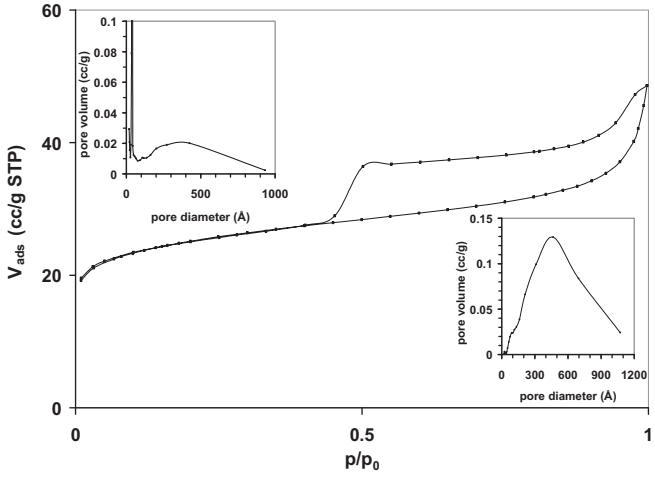


Fig. 4. Nitrogen adsorption isotherm of internal char deposited on Ce–Zr–O and pore volume distribution of this sample before (top inset) and after (bottom inset) an oxidation treatment.

of Ce–Zr–O, in line with the SEM results. An oxidation treatment of this sample resulted in a macropore volume comparable to that of the fresh Ce–Zr–O, confirming the location of char in the Ce–Zr–O pores (inset in Fig. 4 (bottom)).

Fig. 5 presents the XPS spectra of external char, Ce–Zr–O and internal char deposited on Ce–Zr–O in the regions of the binding energy of the C 1s and O 1s. Fig. 5a and b correspond to C 1s spectra of internal char deposited on Ce–Zr–O and external char in the absence of Ce–Zr–O, respectively. Both spectra present a main peak at 285 eV, attributed to C–C and C–H bonds [21–23] in hydrocarbons, and a long tail at higher binding energies which is composed of 4 overlapping individual peaks associated to O–C type bond (C–O, C=O, O–C=O and CO_3) [21–23]. Comparison between the C 1s spectra of both samples (Fig. 5c) clearly shows that O–C type bond (>286 eV) is more dominant in internal char (formed in the presence of the Ce–Zr–O catalyst).

Spectra in the region of the binding energy of the O 1s electrons are depicted in Fig. 5d–g. The spectrum corresponding to Ce–Zr–O (Fig. 5d) presents a peak which can be described as contribution from two different types of oxygen. One type of oxygen is assigned to a main peak centered at 529.6 eV and the second one is associated to a minor peak centered at 531.4 eV. The O 1s fitting of the spectrum corresponding to external char is shown in Fig. 5e. In this case, the optimal fitting of the curve is constituted of 4 peaks, at positions 531.6 eV, 533.7 eV, 536.7 eV and 539.2 eV. The first two peaks are assigned to C=O and C–O bonds, respectively, and the other two at higher binding energies are satellite peaks [21–23]. Fig. 5f shows the O 1s fitting spectrum of internal char deposited on Ce–Zr–O. A broad peak in the range 528–538 eV was observed. The best fitting for this peak was obtained with the addition of 4 individual bands at the following binding energies: 529.9 eV, 531.5 eV, 532.6 eV and 534.17 eV. Fig. 5g shows the overlapping of the O 1s spectrum of internal char deposited on Ce–Zr–O (light solid line) with the individual curves and the resulting fitting of the O 1s spectra corresponding to Ce–Zr–O (dash lines) and external char (dark solid lines). To do so, the parameters associated with the main band for both Ce–Zr–O and external char, namely peak position, area and full width half maximum (FWHM), were constrained when comparing them with the O 1s spectrum of internal char deposited on the catalyst. The parameters associated with the rest of the indi-

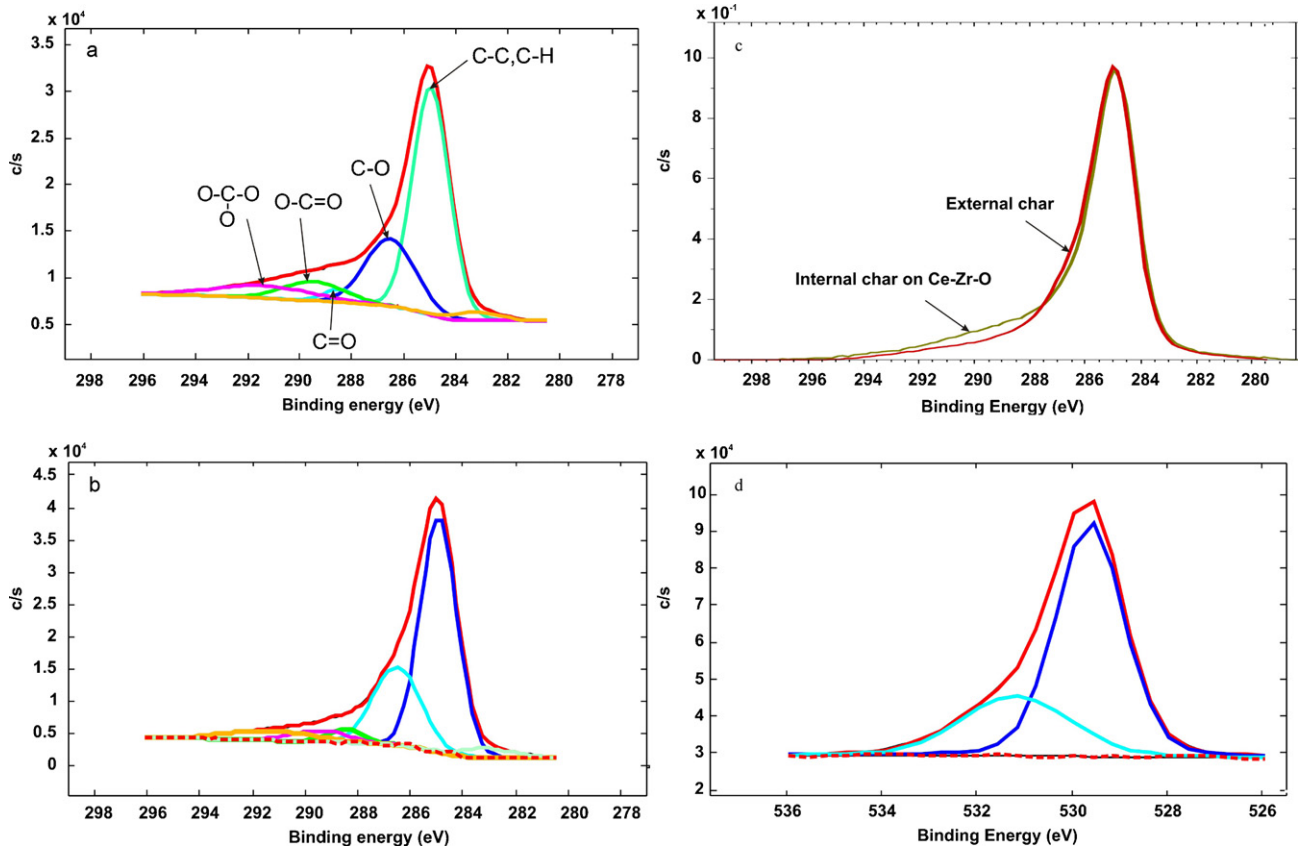


Fig. 5. XPS spectra of (a) C 1s internal char deposited on Ce–Zr–O, (b) C 1s external char, (c) comparison C 1s external char (dark line)/internal char deposited on Ce–Zr–O (light line), (d) O 1s Ce–Zr–O, (e) O 1s external char, (f) O 1s internal char deposited on Ce–Zr–O, and (g) comparison O 1s Ce–Zr–O (dash lines)/external char (dark solid lines)/internal char deposited on Ce–Zr–O (light solid line).

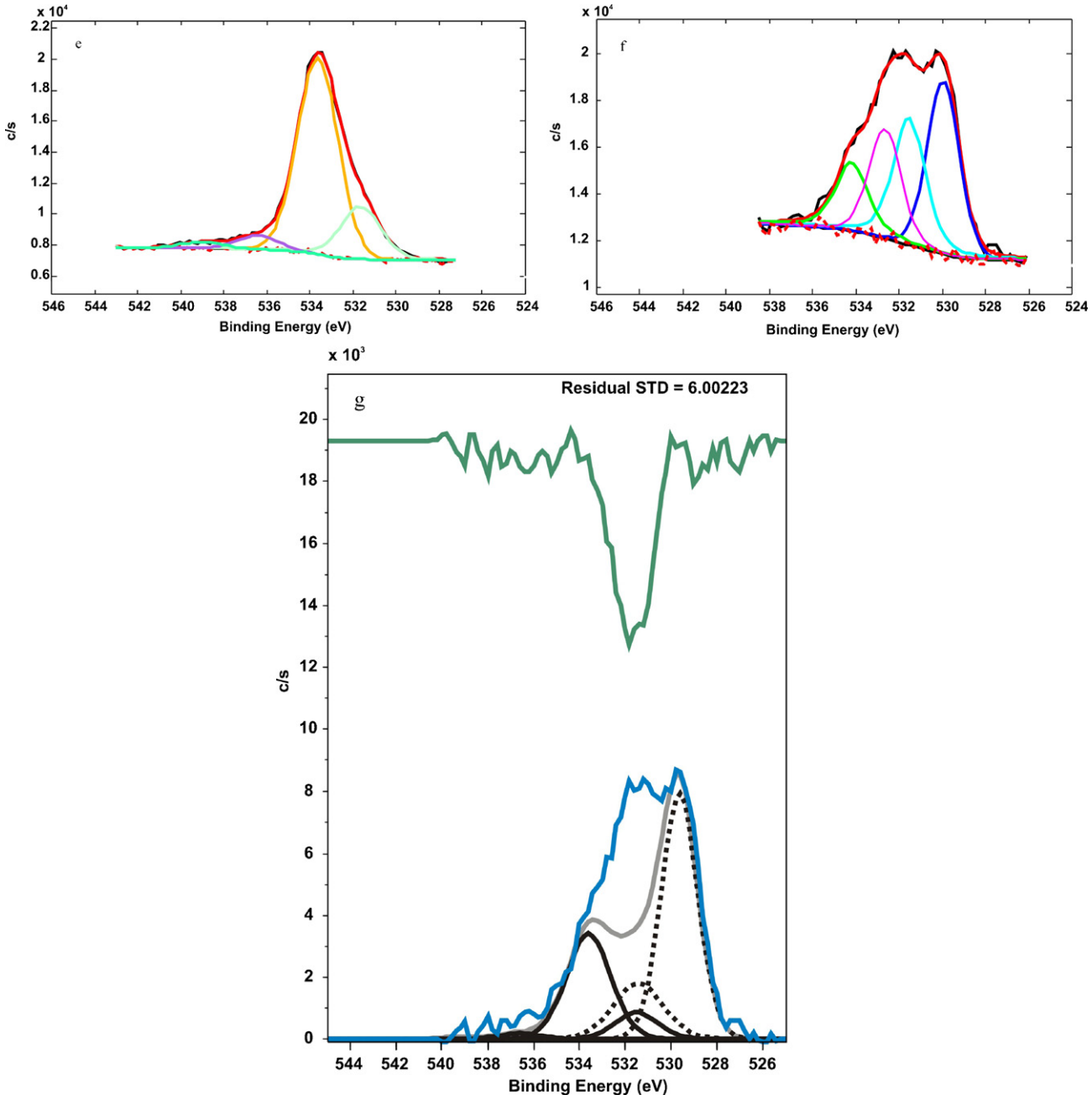


Fig. 5. (*Continued*).

vidual bands depended on those associated with the main band (see Tables 1 and 2). It is clear from Fig. 5g that the fitting resulting from the sum up of the 6 individual bands (2 from Ce–Zr–O and 4 from external char) was not equivalent to the O 1s spectrum associated with internal char deposited on Ce–Zr–O. There was remaining intensity which could not be explained by any of the individual components. This remaining peak, so-called residual, shown on the top of the figure, was in the binding energy range 530.2–533.5 eV. This binding energy range is typical of O–C bonds (CO_3 [21]), in good agreement with the findings obtained for C 1s (Fig. 5a).

3.4. Reactivity measurements

3.4.1. CO_2 gasification

Fig. 6 shows a typical result for CO_2 (dry) gasification of char expressed as char conversion *vs.* reaction time at 700 °C. Reactions rate can be calculated with the derivative. The resulting

Table 1
Quantification parameters of the O 1s spectrum of Ce–Zr–O.

Band	Peak position	Area	FWHM	Delta peak position	Area	Delta FWHM
1	529.6	81,655	1.80	0	*A*	0
2	531.4	25,969	2.54	+1.8	$A \times 0.318$	+0.74

Table 2
Quantification parameters of the O 1s spectrum of char.

Band	Peak position	Area	FWHM	Delta peak position	Area
1	533.7	30,516	2.23	0	A
2	531.6	7714	2.23	−2.1	$A \times 0.252$
3	536.7	1731	2.23	+3.0	$A \times 0.057$
4	539.2	744	2.23	+5.5	$A \times 0.024$

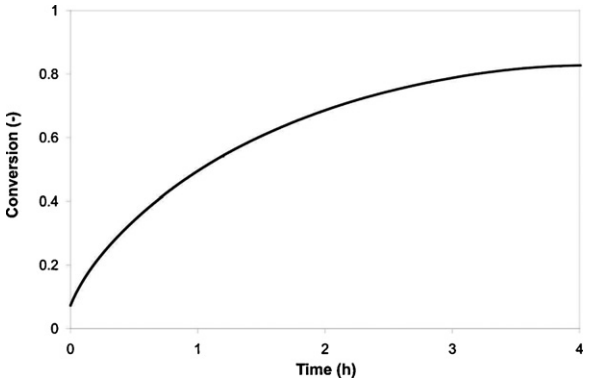


Fig. 6. Char conversion as a function of reaction time for CO_2 gasification of internal char deposited on Ce–Zr–O at 700 °C. CO_2 flow: 20 ml min^{-1}; Ar flow: 40 ml min^{-1}.

rates for internal char (deposited on Ce–Zr–O) in the temperature range between 600 and 800 °C are shown in Fig. 7 as a function of the conversion. As can be clearly seen, char reactivity increased sharply with temperature. The initial rate of gasification at 800 °C was approximately one order of magnitude higher than that at 600 °C. The influence of temperature in the reactivity profile is also observed in Fig. 7. At low temperatures (600 °C), char was gasified at a relatively constant rate throughout the whole experiment whereas the rate of gasification at higher temperatures (700 °C and 800 °C) decreased significantly with conversion.

Comparison of char reactivity at 30% conversion in the presence (internal char) and absence (external char) of Ce–Zr–O is illustrated in Fig. 8. As expected, the presence of Ce–Zr–O had a significant influence on the rate of gasification, especially at the higher temperatures, e.g., the presence of Ce–Zr–O allowed char to gasify one order of magnitude faster than char gasified non-catalytically.

Fig. 8 also shows the reactivity at 30% conversion of internal char deposited on SiO_2 at 700 °C. The reactivity in this case was comparable to that observed non-catalytically, indicating the inertness of SiO_2 and the inability to enhance char gasification.

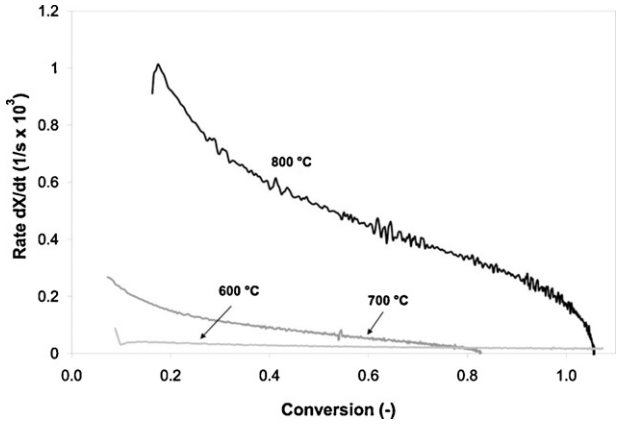


Fig. 7. Reactivity of internal char in the presence of catalyst as a function of the degree of conversion during CO_2 gasification. CO_2 flow: 20 ml min^{-1}; Ar flow: 40 ml min^{-1}. T = 600 °C ( ); T = 700 °C ( ); T = 800 °C ( ). TGA set-up.

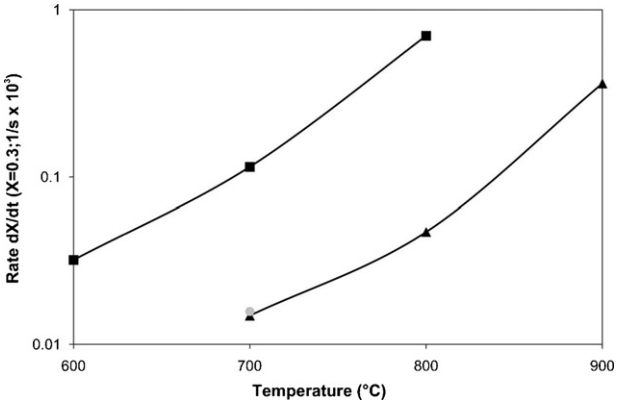


Fig. 8. Char reactivity during CO_2 gasification at 30% conversion as a function of temperature in the absence (external char) and presence (internal char) of catalyst. (▲) external char, (■) internal char deposited on Ce–Zr–O and ( ) internal char deposited on SiO_2. CO_2 flow: 20 ml min^{-1}; Ar flow: 40 ml min^{-1}.

Table 3 compares the activation energies calculated according the Arrhenius equation for the catalytic (internal char deposited on Ce–Zr–O) and the non-catalytic (external char) CO_2 gasification. As expected, the presence of Ce–Zr–O clearly lowered the activation energy and thus increased the reaction rate.

3.4.2. Steam gasification

Reactivity profiles for the steam gasification of internal char deposited on Ce–Zr–O as a function of the degree of conversion and temperature are shown in Fig. 9. Similar to dry (CO_2) gasifi-

Table 3
Activation energies calculated by the Arrhenius equation for the catalytic and non-catalytic steam and CO_2 gasification of char.

	E_a (kJ mol^{-1})	
	CO_2	H_2O
Catalytic	119	157
Non-catalytic	160	274

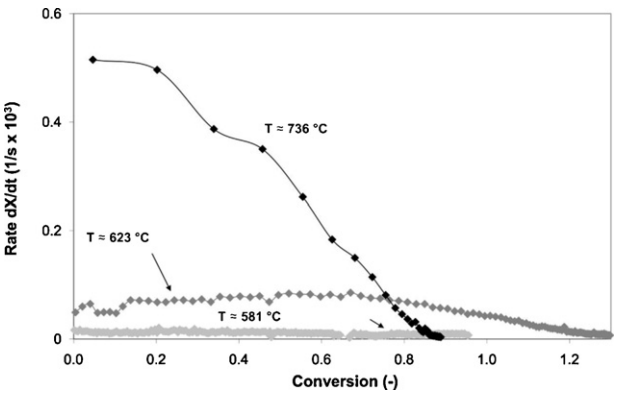


Fig. 9. Reactivity of internal char in the presence of catalyst as a function of the degree of conversion during steam gasification. Steam flow: 0.15–0.5 g min^{-1}, N_2 flow: 9 ml min^{-1} $T \approx 581$ °C ( ), $\Delta p \approx 0.2$ bar; $T \approx 623$ °C, $\Delta p \approx 0.2$ bar ( ); $T \approx 736$ °C, $\Delta p \approx 0.3$ bar ( ). Fixed-bed reactor.

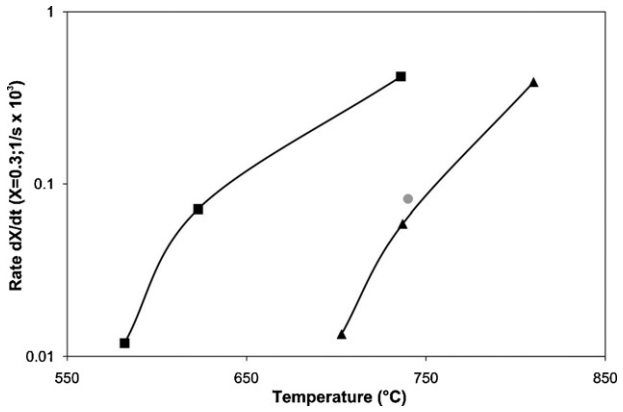


Fig. 10. Char reactivity during steam gasification at 30% conversion as a function of temperature in the absence (external char) and presence (internal and physical mixing char) of catalyst. (▲) external char, (■) internal char deposited on Ce–Zr–O and (●) physical mix char with Ce–Zr–O. Steam flow: 0.15–0.5 g min^{-1}; N_2 flow: 9 ml min^{-1}.

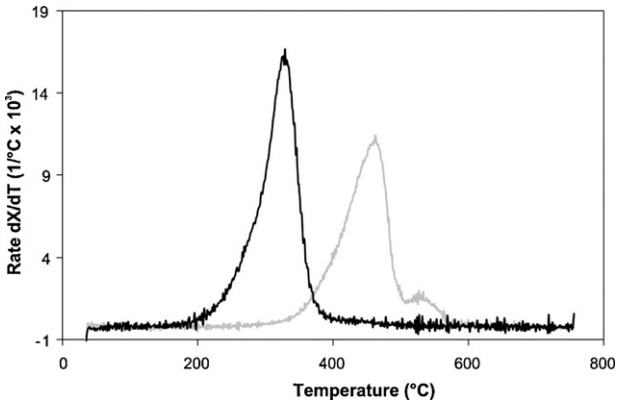


Fig. 11. Char reactivity during combustion as a function of temperature in the absence (external char) and presence (internal char) of catalyst. Air flow: 20 ml min^{-1}; Ar flow: 40 ml min^{-1}; heating rate: 5 °C min^{-1}. Internal char deposited on Ce–Zr–O (▬); external char (▬).

cation, both reactivity profile and rate of gasification were found to strongly depend on temperature. Char gasification at 580 °C was very slow and constant. The increase in temperature up to 625 °C increased the rate of gasification by a factor of 4 and resulted in a comparable reactivity profile. Char reactivity was further increased when carrying out experiments at 735 °C. Reactivity at this reaction temperature was approximately 40 times faster than that obtained at 580 °C. In contrast to the observations at lower temperatures, reactivity at 735 °C decreased as char was being consumed.

Char reactivity for steam gasification in the presence (internal char) and absence (external char) of Ce–Zr–O is compared at 30% conversion in Fig. 10. Similar to our findings for CO_2 gasification, char reactivity during steam gasification was enhanced by the presence of Ce–Zr–O, however to a lower extent. Comparison between reaction times at approximately 735 °C indicates that char gasification occurred ~3.5 times faster when the reaction was carried out catalytically.

Regarding the gas product distribution, H_2, CO and CO_2 were the only products formed during gasification of the internal char deposited on Ce–Zr–O. In the non-catalytic case (external char), besides the above mentioned products, small amounts of CH_4 were also observed in the beginning of the reaction. The CO/CO_2 ratio during catalytic internal char gasification was approximately 0.06 (much more CO_2) whereas CO and CO_2 concentrations were comparable in the non-catalytic case, indicating that in the presence of Ce–Zr–O the water–gas shift equilibrium was completely shifted towards CO_2 and H_2 production.

Fig. 10 also includes reactivity of char physically mixed with Ce–Zr–O (solid–solid contact) at 740 °C. It is obvious from the figure that the reactivity of this char was remarkably slower (~3 times) than that observed for internal char deposited on Ce–Zr–O and comparable to that associated with external char (gasified non-catalytically) under similar reaction conditions. These results clearly indicate that physical mixing of char with catalyst particles resulted in poor contact and thus, char gasification is only enhanced marginally by the presence of the Ce–Zr–O catalyst.

Table 3 shows, similarly to the observations on CO_2 gasification, that the presence of Ce–Zr–O (internal char) decreased the activation energy significantly, thus increasing the rate of steam gasification.

3.4.3. Combustion

Fig. 11 shows the reactivity profiles for char combustion in the presence (internal char) and absence (external char) of Ce–Zr–O as a function of temperature. Char combusted non-catalytically resulted in a main peak centered around 460 °C and a smaller, second peak with maximum at 530 °C. The presence of Ce–Zr–O enhanced char combustion drastically, as evidenced by the presence of a single peak centered at 330 °C.

4. Discussion

This work reports on catalytic steam gasification of pyrolysis-oil derived char and adds on our previous non-catalytic findings. Additionally, this work addresses catalytic CO_2 gasification and char combustion as alternative approaches to convert char formed during pyrolysis oil evaporation. The kinetic results convincingly show the effectiveness of the ceria–zirconia catalyst to convert char to syngas *via* catalytic steam and CO_2 gasification at relatively low temperatures (<700 °C), as evidenced by the substantial enhancement in char reactivity and the decrease in activation energies, as compared to the non-catalytic scenario (see Figs. 8 and 10).

In addition, this catalyst showed excellent combustion activity. In the presence of Ce–Zr–O, internal char was fully combusted at temperatures below 400 °C whereas non-catalytic combustion of (external) char required temperatures up to ~570 °C. This enhanced reactivity by using ceria-based catalyst is in good agreement with results of McKee [24] on combustion of graphite, reporting that addition of Ce to the graphite allowed initial combustion of graphite at 450 °C, instead of 700 °C in absence of catalyst.

It is here appropriate to recall that external char, produced in the absence of catalyst, was formed at 525 °C. On the other hand, internal char, formed *in situ* on the catalyst, was produced by heating the catalyst impregnated with oil up to 750 °C. Thus, the char produced catalytically was formed at higher temperatures. We have recently reported that reactivity of char strongly depends on the temperature at which it is formed [3]. The higher the temperature the more devolatilization (gas release) char undergoes, resulting in a lower reactivity of char. Accordingly, the char produced in the presence of catalyst should be less reactive than char produced non-catalytically. Therefore, the already excellent catalytic effect of the ceria–zirconia catalyst on gasification and combustion rates is even more significant and as a consequence, the temperature required for char conversion could be lowered even further, especially since pyrolysis oil evaporation can be carried out at lower temperatures, e.g., 500 °C.

It is well known that CeO_2 has excellent oxygen exchange capacity involving red-ox changes between Ce^{4+} and Ce^{3+} [25] and that it has ability to activate H_2O and CO_2. Sharma et al. [15] clearly proved that re-oxidation of CeO_{2-x} by CO_2 occurs at temperatures as low as 350 °C. Further, water activation (CeO_{2-x} re-oxidation)

over ceria was reported to proceed rapidly at 550 °C [26]. Therefore both reactants, CO_2 and H_2O, are activated on Ce–Zr–O at the reaction temperatures used in this study. Interestingly, our kinetic results reveal that non-catalytic steam gasification of char was significantly faster (up to one order of magnitude at 800 °C) than CO_2 gasification (compare Figs. 8 and 10). In contrast, under comparable reaction conditions and in the presence of Ce–Zr–O, the rate for steam gasification was only slightly higher than that observed for CO_2 gasification. These facts indicate that the Ce–Zr–O red-ox catalyst is able to speed up gasification with CO_2 even more effectively than with water. Care should be taken for direct comparison between steam and CO_2 gasification activation energies, as different setups and settings and therefore different dynamics have been applied for each type of gasification.

Differences in the CO/CO_2 ratio and H_2 amounts between catalytic and non-catalytic gasification of char confirms that Ce–Zr–O can readily activate both H_2O and CO_2. In the case of catalytic steam gasification (internal char deposited on Ce–Zr–O), the CO/CO_2 ratio was much lower than that observed non-catalytically (external char) (0.06 *vs.* 1), indicating that in the presence of Ce–Zr–O the water–gas shift equilibrium was completely shifted towards CO_2 and H_2 production, in agreement with literature [27–29]. It is important to highlight that, in contrast to alkali-based and earth-alikali based (K, Na, Mg, Ca, etc.) or Ni-based catalysts [30–32], used in catalytic coal gasification, ceria–zirconia catalysts do not catalyze the methanation reaction when mixed with carbon [13,31,33,34] and thus hydrogen yields remain high.

Characterization results suggest that Ce–Zr–O also plays a crucial role in the nature of char formed during pyrolysis oil evaporation (internal char) and this could also explain the differences in char reactivity between catalytic and non-catalytic conditions. It was concluded from XPS spectra that char produced during pyrolysis oil evaporation in the presence of Ce–Zr–O was more oxygenated than that formed non-catalytically. Domazetis et al. [31] observed similar oxygenate species for the catalytic gasification of low-quality coals over Fe-based catalysts. The authors speculated that Fe-oxygenate complexes formed during pyrolysis participate in metal assisted pyrolysis and gasification. As mentioned earlier, it is well established that ceria possesses excellent red-ox properties [35,36], generating oxygen vacancies and thus allowing release and storage of oxygen ("oxygen storage capacity" (OSC)) [36,37]. Therefore, we may speculate that occurrence of oxygen release from ceria during pyrolysis oil evaporation resulted in oxygen vacancies and formation of new carbon–oxygen bonds in char (e.g., carbonyl and carbonate groups). H_2O and CO_2 (potential oxidants) released in gas phase during evaporation would be able to fill in the generated oxygen vacancies in ceria.

Evolution of these new C–O type of bonds suggests that the contact between (internal) char and Ce–Zr–O was excellent. This hypothesis was confirmed by SEM images (Fig. 2), which showed that char was located inside the pores of Ce–Zr–O and thus in intimate contact with the catalyst. Change in porosity between fresh Ce–Zr–O and Ce–Zr–O mixed with internal char (compare Figs. 1 and 4) also suggested that char was mostly deposited in the macropores of Ce–Zr–O, in agreement with our previous findings. Further porosity measurements of a sample containing internal char deposited on Ce–Zr–O which had been previously submitted to an oxidation treatment (O_2/He) (inset in Fig. 4 (bottom)) corroborated the location of char by showing comparable porosity to that observed for the fresh Ce–Zr–O. A small amount of char, however, formed an external thin surface layer of few μm thick (see Fig. 3b). This suggests that a small fraction of the oil did not penetrate into the catalyst pores prior to char formation, resulting in overimpregnation of the catalyst particles. As a consequence, this small amount of char was not in intimate contact with the catalyst, resulting in a slightly lower rate of gasification. Therefore, higher char reactivities could be obtained by optimizing (i) catalyst/pyrolysis ratios and (ii) contact between catalyst/char.

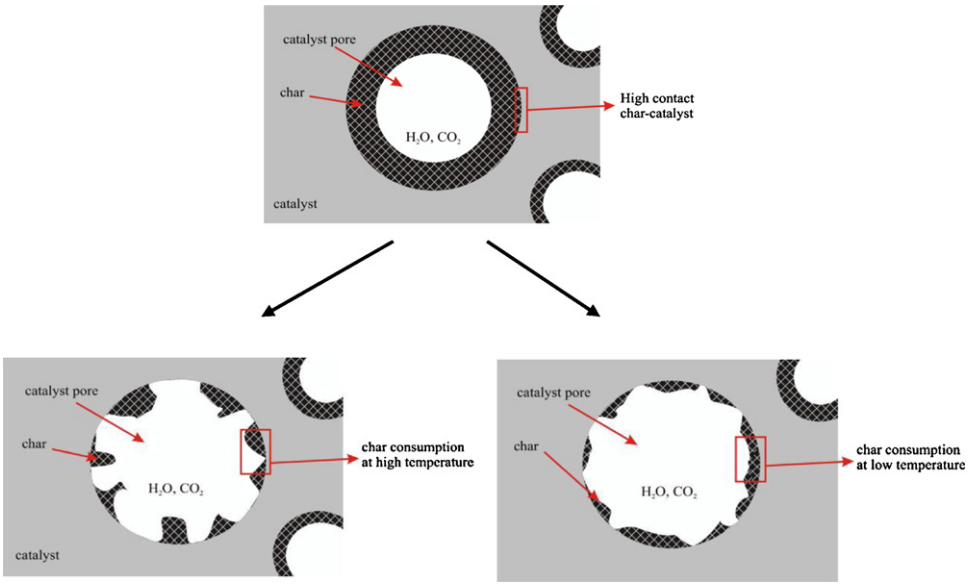
Up to this point, our findings clearly show the significant role of the Ce–Zr–O catalyst in enhancing char reactivity during steam and CO_2 gasification; however, discrimination between the role of the catalyst during pyrolysis oil evaporation and that during char gasification requires further research.

In the case of coal gasification, it has been stated in literature that surface area plays an important role in the rate of gasification. The higher the surface area the faster the gasification rate is [38,39]. In order to study the influence of catalyst surface area in char gasification, CO_2 gasification of internal char was carried out over a SiO_2 catalyst with very high surface area (351 $m^2\ g^{-1}$) and with no ability to activate CO_2. The low gasification rate obtained over this SiO_2 catalyst (Fig. 8), comparable to that observed non-catalytically and much lower than that over Ce–Zr–O, confirmed the essential role of Ce–Zr–O in activating CO_2 and H_2O and thus enhancing char gasification and ruled out any positive contribution of surface area of non-catalytic materials in the rate of char gasification. Based on our observations and discussions we thus propose that ceria provides oxygen for char gasification, resulting in oxygen vacancies, and that activation of CO_2 and H_2O on Ce–Zr–O leads to regeneration of the oxygen vacancies. This is consistent with our studies of steam reforming of acetic acid [16] and water–gas shift [40], where the role of ceria is suggested and observed. Another important observation is that differences in char production which are expected to result in different char structures, yielded to comparable gasification rates, as observed for internal char deposited on SiO_2 (char produced at low heating rate) and external char (char produced at high heating rate [3]).

The excellent catalytic activity of Ce–Zr–O could also explain the dependence of reactivity profiles of internal char on temperature for both steam and CO_2 gasification (Figs. 7 and 9). At low temperatures, the rate of gasification was relatively constant throughout the whole conversion range. In contrast, at high temperatures, the rate of gasification decreased significantly with the degree of conversion. Scheme 1 illustrates the proposed explanation for this behavior. Before reaction (top image) one part of the char is in intimate contact with the Ce–Zr–O particles inside the catalytic pores and therefore will react catalytically. Another part of the char, also inside the catalyst pores, is located at the outer shell of the catalytic surface and therefore will be gasified non-catalytically. Thus, both catalytic and non-catalytic gasification occur simultaneously. Due to the high catalytic activity of Ce–Zr–O, the char in close contact with the catalyst particles will react faster to react away, as it has been shown that catalytic gasification occurs much faster than the analogous non-catalytic. As a result, the catalyst pores will start to retrieve their porosity and the remaining char which will be submitted to forces due to tension will induce break up of the initial char structure. This char will re-contact the catalytic surface leading, however, to a smaller char-catalyst interface. As noticed above, that catalytic gasification is much faster than the analogous non-catalytic and therefore poorer contact between the char and the catalyst will result in a decrease in gasification rate with increasing conversion. As proposed in Scheme 1, char gasified at high temperatures (bottom left image) results in a more abrupt decrease in char-catalyst interface than char gently gasified at lower temperatures (bottom right image), resulting in a decrease in the rate of gasification with increasing the degree of conversion to a larger extent than at lower temperatures.

Based on the obtained results, Scheme 2 illustrates the possible paths involved in steam and CO_2 char gasification in the presence and absence of Ce–Zr–O.

Char gasification occurs mainly *via* two parallel routes. On one hand, the initial char, which is slightly oxygenated, here referred as C, is first converted to more reactive intermediates (further oxy-

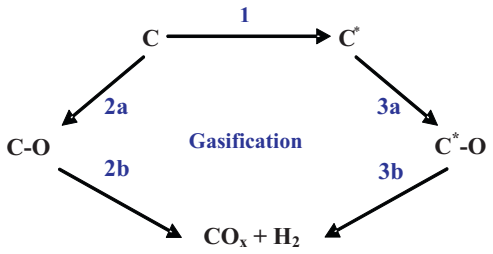


Scheme 1. Proposed explanation for the dependence of char reactivity profiles on temperature. Char deposited on porous Ce–Zr–O before reaction (top) and char consumption at high temperatures (bottom left) and low temperatures (bottom right).

genated char), here denoted as C–O (route 2a), which react further to lead to the end products ($CO_x + H_2$) (route 2b). As discussed earlier, pyrolysis oil evaporation in the presence of Ce–Zr–O resulted in the generation of more oxygenated char (C–O bond type species). Therefore, according to the proposed scheme, Ce–Zr–O facilitates the pre-steps involved in char gasification (route 2a), explaining, to a certain extent, the enhancement in char reactivities.

In previous work on non-catalytic steam gasification [3], we concluded that, similar to coal gasification [41], part of the initial char undergoes a gradual transformation (aging process) towards a hydrogen and oxygen lean type of char, denoted as C*, which can be accompanied by volatilization (gases release). This results in a more stable and thus less reactive char (route 1). It was further concluded that volatilization is favored at high temperatures. In the present study, char on the catalyst was produced at 750 °C. According to the non-catalytic results, at this temperature, volatilization should have already occurred to a large extent. Therefore, further volatilization during catalytic CO_2 and steam gasification is expected to occur only marginally. Similar to the analogous route described for the initial char, the aged char, C*, is further converted to more reactive oxygenate intermediates, represented as C*–O, (route 3a) which are gasified to $CO_x + H_2$ (route 3b). Also in this case, Ce–Zr–O is suggested to enhance the route towards formation of oxygenates, resulting in an enhancement in gasification rates.

To summarize, Ce–Zr–O enhanced CO_2 and steam gasification of char derived from pyrolysis oil significantly. Based on our observations and discussions addressed in this work, we conclude that this enhancement could be further improved. Two main aspects should be considered for optimization: the catalyst/char contact and the temperature in which char is produced on the catalyst.



Scheme 2. Proposed mechanism for CO_2 and steam gasification of char derived from pyrolysis oil.

5. Conclusions

Our findings convincingly demonstrate that pyrolysis-oil evaporation in combination with internal catalytic char gasification is feasible at relatively low temperatures (≤700 °C). This can be explained by the high efficiency of the ceria–zirconia based catalyst to convert char to syngas *via* catalytic steam/CO_2 gasification at those low temperatures. This catalyst can replace the conventional silica used as bed material during pyrolysis oil evaporation, hence, improving the performance of the two-staged reactor concept. The role of the Ce–Zr–O catalyst is suggested to provide oxygen for char gasification, resulting in oxygen vacancies, and activate H_2O and CO_2 to regenerate the oxygen vacancies formed. Additionally, Ce–Zr–O facilitates the pre-steps involved in char gasification, by making char more oxygenated and thus more reactive during pyrolysis-oil evaporation. It has been shown that a good contact between char and catalyst is essential for the catalytic enhancement in char gasification. The catalyst/pyrolysis-oil ratio and the temperature in which char is produced are two aspects to be optimized.

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